

report — MetaGenomics Dataset

Complete aggregated dataset · Scientific Papers + Bioinformatics Forums + GitHub Issues · 29 bioinformatics categories · Full metagenomics pipeline coverage

Prepared by: *youssef lamrabi*

44,415

Total Records

Papers + Forums + GitHub

31,636

Paper Records

Scientific literature · 70.5%

10,253

Forum Records

Bioinformatics QA forums · 23.1%

2,526

GitHub Records

Troubleshooting (issues) · 5.7%

Figure 1 — Record Type Distribution (Complete Dataset)

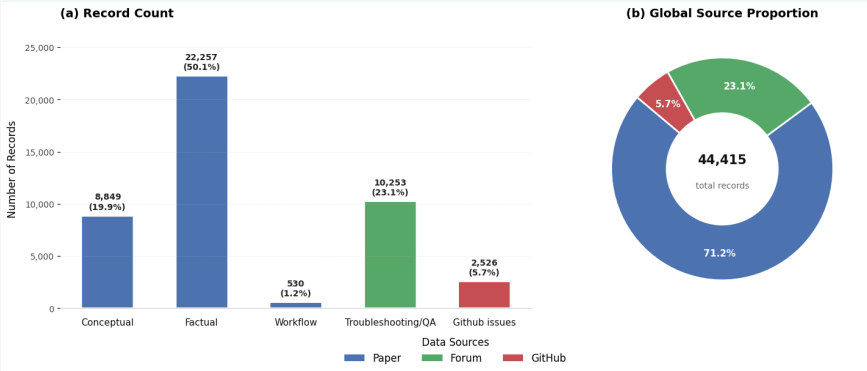
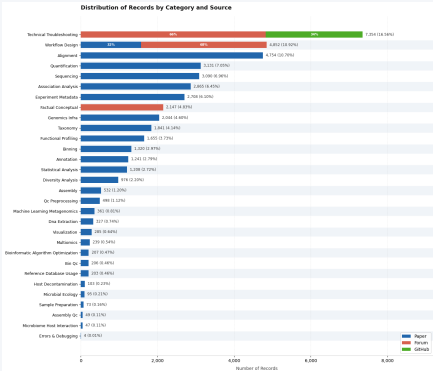


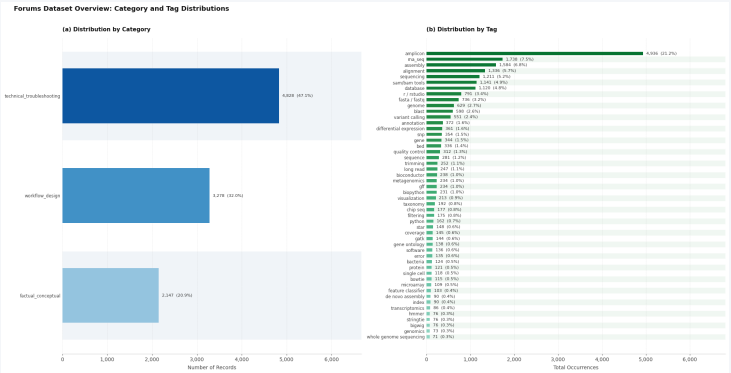
Figure 2 — Category Distribution by Source (Paper / Forum / GitHub)



Unified type composition across Papers + Forums + GitHub (44,415 records). Absolute counts per type. Right: donut proportion chart.

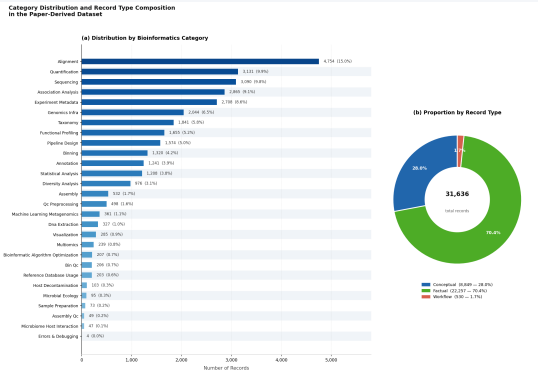
Stacked bars: relative contribution of Papers, Forums, and GitHub per bioinformatics category. Shared categories are merged.

Figure 3 — Forum Dataset: Category & Tag Distribution



10,253 forum records: technical_troubleshooting 47.1%, workflow_design 32.0%, factual_conceptual 20.9%.

Figure 4 — Paper-Derived Dataset: Category & Type Composition



Bioinformatics category distribution (28 categories). Type breakdown — Conceptual 28.0%, Factual 70.4%, Workflow 1.7% — for 31,636 paper records.

Out-of-Scope Detection — Similarity-Based Approaches Evaluation

8 embedding approaches tested · Evaluated on $n = 10$ correctly labelled records

Prepared by: youssef lamrabi

Table 1. Per-Sample Similarity Scores and Rank of True Category

#	True Category	BiEnc Centroid	BiEnc Concat	BiEnc LLM	Cross Encoder	Zero-Shot NLI	BiEnc KW Avg	BioBERT	KeyBERT Matrix
1	alignment	0.2498 rank 22	0.1810 rank 17	0.2806 rank 22	-6.0770 rank 13	0.1160 rank 17	0.2336 rank 22	0.6622 rank 13	0.2723 rank 18
2	alignment	0.8084 rank 2	1.0000 rank 1	1.0000 rank 1	-1.5369 rank 1	0.5124 rank 6	0.7954 rank 4	1.0000 rank 2	0.6358 rank 2
3	pipeline design	0.4105 rank 12	0.2236 rank 16	0.5099 rank 18	-3.9510 rank 5	0.1028 rank 23	0.4072 rank 15	0.6359 rank 9	0.0000 rank 16
4	taxonomy	1.0000 rank 1	0.7681 rank 3	0.8744 rank 1	-2.2454 rank 2	0.0440 rank 15	0.9998 rank 1	0.9425 rank 2	1.0000 rank 1
5	binning	0.2509 rank 8	0.0000 rank 12	0.1722 rank 18	-4.0851 rank 5	0.4831 rank 3	0.2383 rank 8	0.5161 rank 5	0.0894 rank 7
6	binning	0.2436 rank 11	0.0529 rank 10	0.2126 rank 15	-5.4463 rank 7	0.5668 rank 2	0.2312 rank 12	0.2200 rank 12	0.0498 rank 10
7	sequencing	0.5192 rank 10	0.7633 rank 1	0.7807 rank 1	-7.9775 rank 2	0.7239 rank 5	0.4804 rank 12	0.5056 rank 10	0.4469 rank 5
8	assembly	0.6946 rank 6	0.4079 rank 5	0.5713 rank 3	-4.3141 rank 1	0.0807 rank 14	0.6781 rank 6	0.9175 rank 4	0.6738 rank 4
9	alignment	0.0000 rank 25	0.0006 rank 24	0.1629 rank 22	0.3593 rank 16	0.0000 rank 20	0.0000 rank 25	0.4148 rank 19	0.2760 rank 12
10	binning	0.1791 rank 23	0.0016 rank 24	0.0000 rank 25	-3.3275 rank 13	1.0000 rank 1	0.1675 rank 23	0.0002 rank 19	0.0416 rank 15

■ Rank 1 = correct prediction ■ Rank 2 ■ Rank > 2

Table 1. For each of the 10 evaluation samples, the cosine / relevance score of the true category is reported alongside its rank among all clusters. Rank 1 indicates the model correctly identified the true category as the most similar. Cell colour encodes rank quality (green = rank 1, yellow = rank 2, red = rank > 2).

Table 2. Summary Comparison of Similarity Approaches for Out-of-Scope Detection

#	Approach	Model	Cluster Representation	Scoring Method	Mean Score	Std	Min	Max	Acc@1 (%)
1	Bi-Encoder + Centroid	all-MiniLM-L6-v2	Avg. keyword vectors → centroid	Cosine(row, centroid)	0.4356	0.2989	0.0000	1.0000	10%
2	Bi-Encoder + Concat Phrase	all-MiniLM-L6-v2	Keywords concatenated as phrase	Cosine(row, phrase emb.)	0.3399	0.3563	0.0000	1.0000	20%
3	Bi-Encoder + LLM Phrase	all-MiniLM-L6-v2	LLM-generated cluster phrase	Cosine(row, phrase emb.)	0.4565	0.3251	0.0000	1.0000	30%
4	Cross-Encoder	ms-marco-MiniLM-L-6-v2	Keywords concatenated as phrase	Relevance score (raw)	-3.8602	2.2525	-7.9775	0.3593	20%
5	Zero-Shot NLI	nli-deberta-v3-small	Category label + top keywords	Entailment probability	0.3628	0.3249	0.0000	1.0000	10%
6	Bi-Encoder + Keyword Avg	all-MiniLM-L6-v2	Individual keyword embeddings	Mean cosine over keywords	0.4191	0.2938	0.0000	1.0000	10%
7	BioBERT	biobert-base-cased-v1.2	LLM-generated cluster phrase	Cosine similarity	0.5814	0.3058	0.0000	1.0000	0%
8	KeyBERT + Matrix Mean	all-MiniLM-L6-v2	Individual keyword embeddings	Mean of cosine matrix	0.3485	0.3171	0.0000	1.0000	10%

Table 2. Evaluation of 8 similarity-based classification approaches tested on $n = 10$ correctly labelled records. Mean Score: average cosine / relevance score of the correct cluster; Acc@1: percentage of samples where the correct category ranked first. Highlighted rows achieved the highest Acc@1. Std: standard deviation.

+ All scores normalized to [0, 1] via min-max scaling for comparability, except Cross-Encoder which retains its raw relevance scale.